

Pattern $\rightarrow$ Tables

• 1, 14, 40, 79, 131, 196

$+13$ $+26$ $+39$ $+52$ $+65$

(13×1) (13×2) (13×3) (13×4) (13×5)

• 5, 17, 41, 77, 125, 185

$+12$ $+24$ $+36$ $+48$ $+60$

(12×1) (12×2) (12×3) (12×4) (12×5)

Even Numbers Multiple Pattern $\rightarrow$

• 5, 31, 83, 161, 265

$+26$ $+52$ $+78$ $+104$

(13×2) (13×4) (13×6) (13×8)

Odd Numbers Multiple Pattern $\rightarrow$

• 11, 24, 63, 128, 219

$+13$ $+39$ $+65$ $+128$

(13×1) (13×3) (13×5) (13×7)

Prime Numbers Multiple Pattern $\rightarrow$

• 8, 30, 63, 118, 195

$+22$ $+33$ $+55$ $+77$

(11×2) (11×3) (11×5) (11×7)

Frequently Asked Pattern $\rightarrow$ $n(n+1)$

2, 6, 12, 20, 30, 42, 56, 72, 90, --

(1×2) (2×3) (3×4) (4×5) (5×6) (6×7) (7×8) (8×9) (9×10) , ---

Pattern 11 → Square based formula Series

Rule :- Based on n^2 or n^2+n type formulas.

Examples,

• $1, 4, 9, 16, 25, 36, 49, \dots$
 $1^2 \quad 2^2 \quad 3^2 \quad 4^2 \quad 5^2 \quad 6^2 \quad 7^2$

• $196, 256, 324, 400, 484, 576$
 $14^2 \quad 16^2 \quad 18^2 \quad 20^2 \quad 22^2 \quad 24^2$

{Even Squares → Consecutive }

• $2, 6, 12, 20, 30, 42$
 $(1 \times 2) \quad (2 \times 3) \quad (3 \times 4) \quad (4 \times 5) \quad (5 \times 6) \quad (6 \times 7)$

{ n^2+n or $n(n+1)$ Pattern }

Practice Questions:

Q1. ?, 25, 36, 49, 64, 81

(A) 19

(B) 15

(C) 11

☒ (D) 16

(E) 21

$(?) , 25, 36, 49, 64, 81$
 $\downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
 $16 \leftarrow 4^2 \quad 5^2 \quad 6^2 \quad 7^2 \quad 8^2 \quad 9^2$

Q2. 20, 30, 42, 56, 72, ?

(A) 97

(B) 89

(C) 92

(D) 100

☒ (E) 90

$20, 30, 42, 56, 72, (?) \rightarrow 90$
 $(4 \times 5) \quad (5 \times 6) \quad (6 \times 7) \quad (7 \times 8) \quad (8 \times 9) \quad (9 \times 10)$

Q3. 41, 54, 69, 86, 105, ?

(A) 138

(B) 124

(C) 123

(D) 126

(E) 125

$41, 54, 69, 86, 105, (?) \rightarrow 105 + 21 = 126$
 $+13 \quad +15 \quad +17 \quad +19 \quad +21$
 OR

(Odd Consecutive No.)

$41, 54, 69, 86, 105, (?) \rightarrow 126$
 $\downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
 $(36+5) \quad (49+5) \quad (64+5) \quad (81+5) \quad (100+5) \quad (121+5)$
 $6^2 \quad 7^2 \quad 8^2 \quad 9^2 \quad 10^2 \quad 11^2$

{ n^2+5 Series }

Pattern 12 → Fibonacci Series *

Rule :- Each term is the sum of previous two terms.

Examples,

$$\begin{array}{ccccccccc}
 2 & , & 3 & , & 5 & , & 8 & , & 13 & , & 21 & , & 34 \\
 & & & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
 & & & & 2+3 & & 3+5 & & 5+8 & & 8+13 & & 13+21
 \end{array}$$

$$\begin{array}{ccccccccc}
 5 & , & 7 & , & 12 & , & 19 & , & 31 & , & 50 \\
 \swarrow & & \searrow & & \swarrow & & \searrow & & \swarrow & & \searrow \\
 +2 & & +5 & & +7 & & +12 & & +19
 \end{array}$$

{ Addition of previous two terms }

$$\begin{array}{ccccccccc}
 8 & , & 10 & , & 18 & , & 28 & , & 46 & , & 74 \\
 & & & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
 & & & & 10+8 & & 18+10 & & 28+18 & & 46+28
 \end{array}$$

Practice Questions:

Q1. 12, 8, 20, 28, 48, ?

(A) 86

(B) 91

(C) 76

(D) 66

(E) 80

$$\begin{array}{ccccccccc}
 12 & , & 8 & , & 20 & , & 28 & , & 48 & , & ? & \rightarrow & 76 \\
 & & & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
 & & & & (12+8) & & (8+20) & & (20+28) & & (28+48)
 \end{array}$$

Q2. ?, 6, 17, 23, 40, 63

(A) 13

(B) 15

(C) 11

(D) 16

(E) 6

$$\begin{array}{ccccccccc}
 11 & , & ? & , & 6 & , & 17 & , & 23 & , & 40 & , & 63 \\
 & & & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
 & & & & (11+6) & & (6+17) & & (17+23) & & (23+40)
 \end{array}$$

Q3. 7, 3, 10, 13, 23, ?

(A) 34

☒ (B) 36

(C) 35

(D) 31

(E) 32

$$\begin{array}{ccccccc}
 7 & , & 3 & , & 10 & , & 13 & , & 23 & , & ? & \rightarrow & 13+23 \\
 & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & & & = 36 \\
 & & (7+3) & & (3+10) & & (10+13) & & (13+23) & & & &
 \end{array}$$

Q4. 4, 11, 15, 26, ?, 67

(A) 40

(B) 43

(C) 37

☒ (D) 41

(E) 42

$$\begin{array}{ccccccc}
 4 & , & 11 & , & 15 & , & 26 & , & ? & , & 67 \\
 & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
 & & (4+11) & & (11+15) & & (15+26) & & (26+?) & & 15+26 \\
 & & & & & & & & & & = 41
 \end{array}$$

Pattern 13 → Difference Based

{ Alternate Square & Cube Differences }

Rule :- Alternate Square and Cube Differences with consecutive base numbers.Examples →

$$\begin{array}{ccccccc}
 9 & , & 13 & , & 40 & , & 56 & , & 181 & , & 217 \\
 \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\
 +4 & +27 & +16 & +125 & +36 & & & & & & \\
 (2^2) & (3^3) & (4^2) & (5^3) & (6^2) & & & & & &
 \end{array}$$

$$\begin{array}{ccccccc}
 1012 & , & 500 & , & 451 & , & 235 & , & 210 & , & 146 \\
 \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\
 -512 & -49 & -216 & -25 & -64 & & & & & & \\
 -(8^3) & -(7^2) & -(6^3) & -(5^2) & -(4^3) & & & & & &
 \end{array}$$

Practice Questions:

Q1. 324, 320, 293, 277, 152, ?

(A) 121

(B) 123

(C) 109

(D) 118

~~(E) 116~~

$$\begin{array}{cccccc}
 324 & , & 320 & , & 293 & , & 277 & , & 152 & , & (?) \\
 \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
 -4 & & -27 & & -16 & & -125 & & -36 & & \\
 -(2^2) & & -(3^3) & & -(4^2) & & -(5^3) & & -(6^2) & &
 \end{array}$$

$152 - 36 = 116$

Q2. 50, 59, 123, 148, ?, 413

(A) 372

(B) 354

(C) 389

~~(D) 364~~

(E) 343

$$\begin{array}{cccccc}
 50 & , & 59 & , & 123 & , & 148 & , & (?) & , & 413 \\
 \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
 +9 & & +64 & & +25 & & +216 & & +49 & & \\
 (3^2) & & (4^3) & & (5^2) & & (6^3) & & (7^2) & &
 \end{array}$$

$148 + 216 = 364$

Q3. 1200, 1175, 959, ?, 398, 317

(A) 925

~~(B) 910~~

(C) 899

(D) 936

(E) 884

$$\begin{array}{cccccc}
 1200 & , & 1175 & , & 959 & , & (?) & , & 398 & , & 317 \\
 \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
 -25 & & -216 & & -49 & & -512 & & -81 & & \\
 -(5^2) & & -(6^3) & & -(7^2) & & -(8^3) & & -(9^2) & &
 \end{array}$$

$959 - 49 = 910$

Q4. 4000, 4169, ?, 2562, 1562, 1643

(A) 2408

(B) 2505

(C) 2415

~~(D) 2441~~

(E)

$$\begin{array}{cccccc}
 4000 & , & 4169 & , & (?) & , & 2562 & , & 1562 & , & 1643 \\
 \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
 +169 & & -1728 & & +121 & & -1000 & & +81 & & \\
 +(13^2) & & -(12^3) & & +(11^2) & & -(10^3) & & +(9^2) & &
 \end{array}$$

$4169 - 1728 = 2441$

Q5. 50, 393, 329, ?, 958, 2289

(A) 1158

(B) 1098

(C) 1058

(D) 1038

(E)

1078

$$329 + 729 = 1058$$

$$\begin{array}{ccccccccc}
 50 & , & 393 & , & 329 & , & ? & , & 958 & , & 2289 \\
 \swarrow & & \searrow & & \swarrow & & \searrow & & \swarrow & & \searrow \\
 +343 & -64 & +729 & -100 & +1331 \\
 + (7^3) & - (8^2) & + (9^3) & - (10^2) & + (11^3)
 \end{array}$$

Q6: 800, 796, 732, 696, ?, 84

(A) 216

(B) 209

(C) 100

(D) 176

(E) 184

$$\begin{array}{ccccccccc}
 800 & , & 796 & , & 732 & , & 696 & , & ? & , & 84 \\
 \swarrow & & \searrow & & \swarrow & & \searrow & & \swarrow & & \searrow \\
 -4 & -64 & -36 & -512 & -100 \\
 (2^2) & (4^3) & (6^2) & (8^3) & (10^2)
 \end{array}$$

$$696 - 512 = 184$$

Pattern 14 :- Constant Ratio { Geometric Progression }

Rule :- Multiply / Divide by a fixed ratio

Examples,

$$\begin{array}{ccccccc}
 3 & , & 12 & , & 48 & , & 192 & , & 768 \\
 \swarrow & & \searrow & & \swarrow & & \searrow & & \swarrow \\
 \times 4 & \times 4 & \times 4 & \times 4
 \end{array}$$

$$\begin{array}{ccccccc}
 160 & , & 80 & , & 40 & , & 20 & , & 10 \\
 \swarrow & & \searrow & & \swarrow & & \searrow & & \swarrow \\
 \div 2 & \div 2 & \div 2 & \div 2
 \end{array}$$

Practice Questions:

Q1. 2916, 972, 324, 108, 36, ?

(A) 13

(B) 15

☒ (C) 12

(D) 14

(E) 9

$$2916, 972, 324, 108, 36, ?$$

$\div 3$ $\div 3$ $\div 3$ $\div 3$ $\div 3$

$36 \div 3 = 12$

Q2. 2, 8, 32, 128, ?, 2048

☒ (A) 512

(B) 472

(C) 522

(D) 509

(E) 513

$$2, 8, 32, 128, ?, 2048$$

$\times 4$ $\times 4$ $\times 4$ $\times 4$ $\times 4$

$128 \times 4 = 512$

Q3. 1127357, 102487, 9317, ?, 77, 7

(A) 837

☒ (B) 847

(C) 857

(D) 867

(E) 877

$$1127357, 102487, 9317, ?, 77, 7$$

$\times 11$ $\times 11$ $\times 11$ $\times 11$ $\times 11$

$77 \times 11 = 847$

Q4. ?, 54, 324, 1944, 11664, 69984

(A) 6

(B) 7

(C) 8

☒ (D) 9

(E) 10

$$?, 54, 324, 1944, 11664, 69984$$

$\times 6$ $\times 6$ $\times 6$ $\times 6$ $\times 6$

$x \times 6 = 54$
 $x = \frac{54}{6}$
 $x = 9$

Pattern 15 :- Variable Ratio (Multiplier Pattern)

Rule :- Multiply by changing numbers { often consecutive }

Examples,

$$\begin{array}{ccccccc} 2 & , & 6 & , & 24 & , & 120 & , & 720 & , & 5040 \\ \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\ \times 3 & & \times 4 & & \times 5 & & \times 6 & & \times 7 \end{array}$$

{ Consecutive Multiples }

$$\begin{array}{ccccccc} 1 & , & 2 & , & 8 & , & 48 & , & 384 & , & 3840 \\ \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\ \times 2 & & \times 4 & & \times 6 & & \times 8 & & \times 10 \end{array}$$

{ Even Consecutive Multiples }

Practice Questions:

Q1. 5, 15, 60, 300, 1800, ?

(A) 12600

(B) 12500

(C) 12598

(D) 12590

(E) 13100

$$\begin{array}{ccccccc} 5 & , & 15 & , & 60 & , & 300 & , & 1800 & , & (?) \\ \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\ \times 3 & & \times 4 & & \times 5 & & \times 6 & & \times 7 \end{array} \quad \begin{array}{l} 1800 \times 7 \\ = 12600 \end{array}$$

Q2. 7, 14, 56, 336, 2688, ?

(A) 26880

(B) 26800

(C) 32256

(D) 28860

(E) 24192

$$\begin{array}{ccccccc} 7 & , & 14 & , & 56 & , & 336 & , & 2688 & , & (?) \\ \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\ \times 2 & & \times 4 & & \times 6 & & \times 8 & & \times 10 \end{array} \quad \begin{array}{l} 2688 \times 10 \\ = 26880 \end{array}$$

Q3. 2, 6, 30, 210, 1890, ?

(A) 22400

(B) 18240

(C) 20790

(D) 24560

(E) None of these

$$\begin{array}{ccccccc}
 2, & 6, & 30, & 210, & 1890, & ? & \\
 \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & & \\
 \times 3 & \times 5 & \times 7 & \times 9 & \times 11 & &
 \end{array}$$

$1890 \times 11 = 20790$

Q4. 5, 10, 60, 600, ?, 151200

(A) 7200

(B) 8000

(C) 8400

(D) 9600

(E) 10800

$$\begin{array}{ccccccc}
 5, & 10, & 60, & 600, & ? & , & 151200 \\
 \downarrow & \downarrow & \downarrow & \downarrow & & & \\
 \times 2 & \times 6 & \times 10 & \times 14 & & & \times 18 \\
 \downarrow & \downarrow & \downarrow & \downarrow & & & \\
 4 & 4 & 4 & 4 & & &
 \end{array}$$

$600 \times 14 = 8400$

Pattern 16 → Progressive Division Pattern

Rule :- Divide by changing numbers {often consecutive}

Examples,

$$\begin{array}{ccccccc}
 720, & 240, & 60, & 12, & 2 \\
 \downarrow & \downarrow & \downarrow & \downarrow & \\
 \div 3 & \div 4 & \div 5 & \div 6 &
 \end{array}$$

{Consecutive Division}

$$\begin{array}{ccccccc}
 7200, & 3600, & 1200, & 300, & 60, & 10 \\
 \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \\
 \div 2 & \div 3 & \div 4 & \div 5 & \div 6 &
 \end{array}$$

Practice Questions:

Q1. 20160, 5040, 1008, 168, 24, ?

(A) 5

(B) 6

(C) 1

~~(D) 3~~

(E) 2

$$20160, 5040, 1008, 168, 24, ? \rightarrow 24 \div 8 = 3$$

$$\begin{array}{cccccc} \div 4 & \div 5 & \div 6 & \div 7 & \div 8 & \end{array}$$

Q2. 10395, 3465, 693, ?, 11, 1

(A) 77

~~(B) 99~~

(C) 55

(D) 66

(E) 88

$$10395, 3465, 693, ?, 11, 1$$

$$\begin{array}{ccccc} \div 3 & \div 5 & \div 7 & \div 9 & \div 11 \end{array}$$

$$693 \div 7 = 99$$

Q3. 92160, 46080, 11520, ?, 240, 24

(A) 2160

~~(B) 1920~~

(C) 1440

(D) 2880

(E) 1680

$$92160, 46080, 11520, ?, 240, 24$$

$$\begin{array}{ccccc} \div 2 & \div 4 & \div 6 & \div 8 & \div 10 \end{array}$$

$$11520 \div 6 = 1920$$

Q4. 76800, 7680, 960, 160, ?, 20

(A) 60

(B) 50

(C) 40

(D) 30

(E) 45

$$76800, 7680, 960, 160, ?, 20$$

$$\begin{array}{ccccc} \div 10 & \div 8 & \div 6 & \div 4 & \div 2 \end{array}$$

$$160 \div 4 = 40$$

Q5. 10395, 945, 105, 15, ?, 1

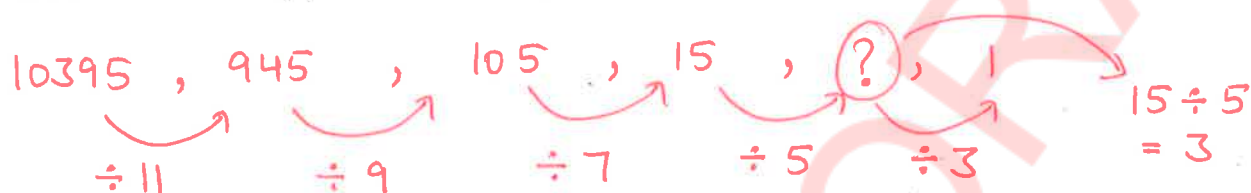
(A) 2

(B) 3

(C) 4

(D) 5

(E) 6



Q6. 1228800, 307200, 38400, ?, 200, 10

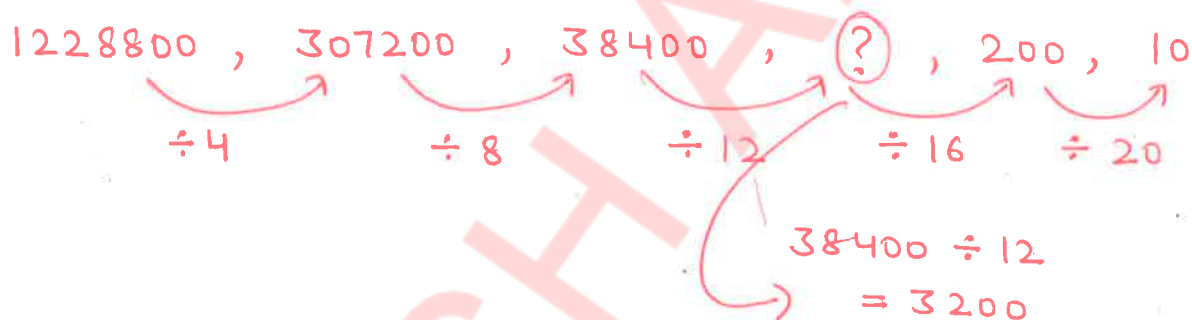
(A) 2400

(B) 2000

(C) 3200

(D) 2800

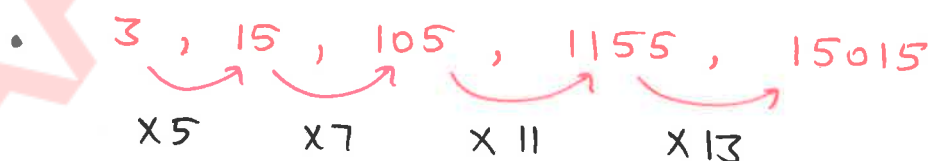
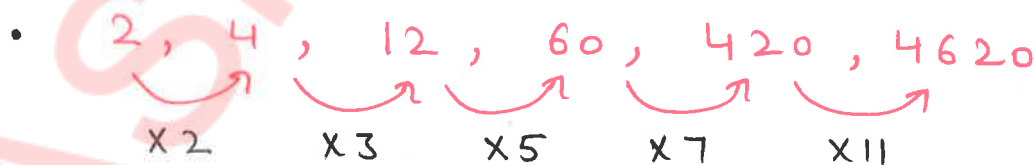
(E) 1600



Pattern 17 → Prime Multipliers

Rule:- Multiply by consecutive prime Numbers.

Examples,



{ Consecutive Prime Multipliers }

Practice Questions:

Q1. 1, 2, 6, 30, ?, 2310

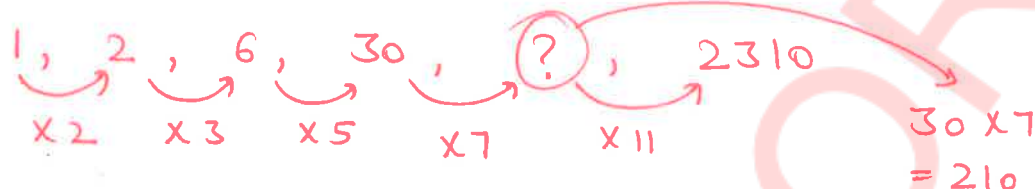
(A) 203

(B) 209

(C) 180

(D) 235

(E) 210



Q2. 3, 9, 45, 315, ?, 45045

(A) 3505

(B) 3467

(C) 3455

(D) 3453

(E) 3465



Q3. ?, 28, 308, 4004, 68068, 1293292

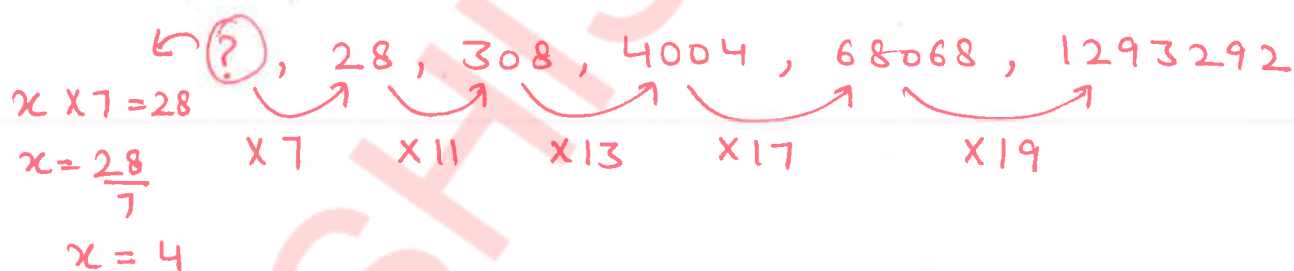
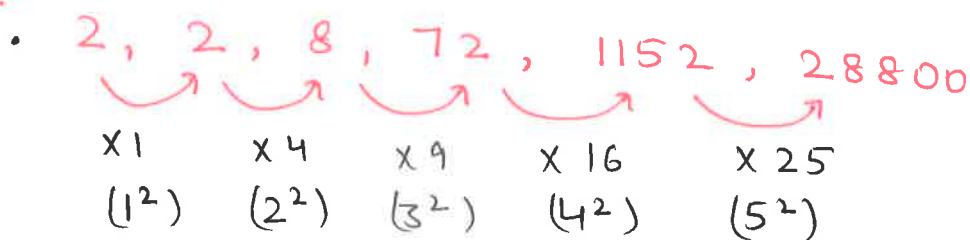
(A) 6

(B) 5

(C) 8

(D) 4

(E) 7

Pattern 18 :- Square MultipliersRule :- Multiply by consecutive Squares ($1^2, 2^2, 3^2, \dots$)Examples,

- $1, 4, 36, 576, 14400, 518400$
 $\times 4 \quad \times 9 \quad \times 16 \quad \times 25 \quad \times 36$
 $(2^2) \quad (3^2) \quad (4^2) \quad (5^2) \quad (6^2)$
 { Consecutive Squares Multiple }

- $6, 216, 10584, 677376$
 $\times 36 \quad \times 49 \quad \times 64$
 $(6^2) \quad (7^2) \quad (8^2)$

Practice Questions:

Q1. 2, 2, 8, 72, 1152, ?

(A) 28700

(B) 28775

(C) 28750

(D) 29300

~~(E)~~ 28800

2, 2, 8, 72, 1152, (?) $\rightarrow$ 1152 $\times$ 25
 $\times 1$ $\times 4$ $\times 9$ $\times 16$ $\times 25$
 (1^2) (2^2) (3^2) (4^2) (5^2)
 $= 28800$

Q2. 3, 12, 108, 1728, ?, 1555200

(A) 43190

(B) 43200

(C) 43210

(D) 43196

(E) 43500

3, 12, 108, 1728, (?), 1555200

$\times 4$ $\times 9$ $\times 16$ $\times 25$ $\times 36$

$1728 \times 25 = 43200$

Q3. 5, 20, 320, ?, 737280, 73728000

(A) 11280

~~(B) 11520~~

(C) 11360

(D) 11480

(E) 11600

5, 20, 320, (?), 737280, 73728000

$\times 4$ $\times 16$ $\times 36$ $\times 64$ $\times 100$

(2^2) (4^2) (6^2) (8^2) (10^2)

Pattern 19 :- Percentage Multipliers

Rule :- Percentage Pattern $\rightarrow$ Multiplier

Examples,

• 10000 , 9000 , 7200 , 5040 , 3024 , 1512
 $\xrightarrow{\times 90\%}$ $\xrightarrow{\times 80\%}$ $\xrightarrow{\times 70\%}$ $\xrightarrow{\times 60\%}$ $\xrightarrow{\times 50\%}$

• 320 , 80 , 40 , 30 , 30 , 37.5
 $\xrightarrow{\times 25\%}$ $\xrightarrow{\times 50\%}$ $\xrightarrow{\times 75\%}$ $\xrightarrow{\times 100\%}$ $\xrightarrow{\times 125\%}$

{ Multiply by Percentage Pattern }

Practice Questions:

Q1. 2500, 2250, 1800, 1260, 756, ?

(A) 372

(B) 384

(C) 378

(D) 396

(E) 360

2500 , 2250 , 1800 , 1260 , 756 , ?
 $\xrightarrow{\times 90\%}$ $\xrightarrow{\times 80\%}$ $\xrightarrow{\times 70\%}$ $\xrightarrow{\times 60\%}$ $\xrightarrow{\times 50\%}$ $\xrightarrow{756 \times 50\%}$
 $= 756 \times \frac{1}{2}$
 $= 378$

Q2. ?, 96, 48, 36, 36, 45

(A) 360

(B) 384

(C) 396

(D) 432

(E) 480

? , 96 , 48 , 36 , 36 , 45
 $\xrightarrow{\times 25\%}$ $\xrightarrow{\times 50\%}$ $\xrightarrow{\times 75\%}$ $\xrightarrow{\times 100\%}$ $\xrightarrow{\times 125\%}$
 $(\times \frac{1}{4})$ $(\times \frac{2}{4})$ $(\times \frac{3}{4})$ $(\times \frac{4}{4})$ $(\times \frac{5}{4})$
 $x \times \frac{1}{4} = 96$
 $x = 384$

Q3. 1250, 500, 300, 240, 240, ?

(A) 264

☒ (B) 288

(C) 120

(D) 320

(E) 360

$$\begin{array}{ccccccccc}
 1250 & , & 500 & , & 300 & , & 240 & , & 240 & , & (?) \\
 \swarrow & & \swarrow & & \swarrow & & \swarrow & & \swarrow & & \swarrow \\
 \times 40\% & & \times 60\% & & \times 80\% & & \times 100\% & & \times 120\% & &
 \end{array}$$

$$\begin{aligned}
 & \rightarrow 240 \times 120\% \\
 & = 240 \times \frac{6}{5} \\
 & = 288
 \end{aligned}$$

Pattern 20 :- $\times a \pm b$ (Fixed Pattern)

Rule :- Multiply by a fixed number and add/subtract a fixed number.

Patterns like $\rightarrow$

$$\begin{array}{l}
 \times 1 + 1, \times 1 + 1, \times 1 + 1, \times 1 + 1 \\
 \times 2 + 2, \times 2 + 2, \times 2 + 2, \times 2 + 2 \\
 \times 3 - 2, \times 3 - 2, \times 3 - 2, \times 3 - 2 \\
 \times 4 - 1, \times 4 - 1, \times 4 - 1, \times 4 - 1
 \end{array}$$

Examples,

$$\begin{array}{ccccccccc}
 2 & , & 5 & , & 14 & , & 41 & , & 122 & , & 365 \\
 \swarrow & & \swarrow & & \swarrow & & \swarrow & & \swarrow & & \swarrow \\
 \times 3 - 1 & & \times 3 - 1 & & \times 3 - 1 & & \times 3 - 1 & & \times 3 - 1 & & \times 3 - 1
 \end{array}$$

$$\begin{array}{ccccccccc}
 3 & , & 8 & , & 18 & , & 38 & , & 78 & , & 158 \\
 \swarrow & & \swarrow & & \swarrow & & \swarrow & & \swarrow & & \swarrow \\
 (\times 2 + 2) & & (\times 2 + 2) & & (\times 2 + 2) & & (\times 2 + 2) & & (\times 2 + 2) & & (\times 2 + 2)
 \end{array}$$

Practice Questions:

Q1. 5, 6, 8, 12, 20, ?

(A) 35

~~(B) 36~~

(C) 34

(D) 40

(E) 41

$$\begin{array}{ccccccccc}
 5 & , & 6 & , & 8 & , & 12 & , & 20 & , & (?) & \rightarrow & (20 \times 2) - 4 \\
 \swarrow & & \swarrow & & \swarrow & & \swarrow & & \swarrow & & \swarrow & & 40 - 4 \\
 \times 2 & & \times 2 & & \times 2 & & \times 2 & & \times 2 & & \times 2 & & 36 \\
 - 4 & & - 4 & & - 4 & & - 4 & & - 4 & & - 4 & &
 \end{array}$$

Q2. 1, 3, 11, 43, ?, 683

(A) 178

~~(B) 171~~

(C) 168

(D) 175

(E) 176

$$\begin{array}{ccccccccc}
 1 & , & 3 & , & 11 & , & 43 & , & (?) & , & 683 \\
 \swarrow & & \swarrow & & \swarrow & & \swarrow & & \swarrow & & \swarrow \\
 \times 4 & & \times 4 & & \times 4 & & \times 4 & & \times 4 & & \times 4 \\
 - 1 & & - 1 & & - 1 & & - 1 & & - 1 & & - 1 \\
 & & & & & & & & & & (43 \times 4) - 1 \\
 & & & & & & & & & & 172 - 1 \\
 & & & & & & & & & & = 171
 \end{array}$$

Q3. 7, 24, 75, 228, ?, 2064

~~(A) 687~~

(B) 677

(C) 697

(D) 707

(E) 702

$$\begin{array}{ccccccccc}
 7 & , & 24 & , & 75 & , & 228 & , & (?) & , & 2064 \\
 \swarrow & & \swarrow & & \swarrow & & \swarrow & & \swarrow & & \swarrow \\
 \times 3 & & \times 3 & & \times 3 & & \times 3 & & \times 3 & & \times 3 \\
 + 3 & & + 3 & & + 3 & & + 3 & & + 3 & & + 3 \\
 & & & & & & & & & & (228 \times 3) + 3 \\
 & & & & & & & & & & 684 + 3 \\
 & & & & & & & & & & \underline{\underline{687}}
 \end{array}$$